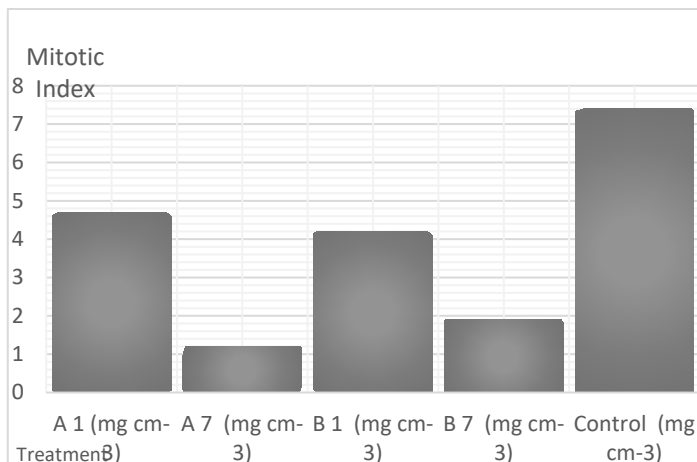


Question Number	Answer	Additional Guidance	Mark
2(a)	An answer that includes the following points : • cut root tip • (root tips) placed in (warm) acid (1) • (root tips) then placed in named stain (1) • (root tip placed on a microscope slide and) {macerated / teased / described} / squashed (1) • use of high power (on a microscope) (1) • correct ref to safety issue (1)	e.g. (acetic / ethanoic) orcein / toluidine blue / methylene blue e.g. rinsing tips in water / wearing gloves / goggles	(5)

Question Number	Answer	Additional Guidance	Mark
2(b)	An answer that includes the following steps: • correct total cell count and total dividing cell count (1) • division of dividing cell count by total cell count (and multiply by 100) (1)	NO ecf e.g. $40 / 4$ ACCEPT 3 ACCEPT any figure in range 35 to 46 e.g. $(4 \div 40) = 0.10$ ($\times 100 = 10.0\%$)	(2)

Question Number	Answer	Additional Guidance	Mark												
2(c)(i)	A graph with the following features : • A axes correct (x – appropriate to their graph , y – mitotic index) and y axis with no break in the axis (1) • L axes correctly labelled with units (1) • P correct plotting on a linear scale on y (1) • S bar chart (1)	e.g.  <table><caption>Mitotic Index Data</caption><thead><tr><th>Treatment</th><th>Mitotic Index</th></tr></thead><tbody><tr><td>A 1 (mg cm⁻³)</td><td>4.8</td></tr><tr><td>A 7 (mg cm⁻³)</td><td>1.2</td></tr><tr><td>B 1 (mg cm⁻³)</td><td>4.2</td></tr><tr><td>B 7 (mg cm⁻³)</td><td>1.8</td></tr><tr><td>Control (mg cm⁻³)</td><td>7.5</td></tr></tbody></table>	Treatment	Mitotic Index	A 1 (mg cm ⁻³)	4.8	A 7 (mg cm ⁻³)	1.2	B 1 (mg cm ⁻³)	4.2	B 7 (mg cm ⁻³)	1.8	Control (mg cm ⁻³)	7.5	(4)
Treatment	Mitotic Index														
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B 1 (mg cm ⁻³)	4.2														
B 7 (mg cm ⁻³)	1.8														
Control (mg cm ⁻³)	7.5														

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	An answer that includes the following points : • each treatment should be repeated (1) • {all conditions / named condition} should be kept constant (1) • (mean and) SD calculated / error bars (1) • look for overlap in SDs / perform t-test (1)	IGNORE standard conditions ALLOW range bars as ecf	(4)